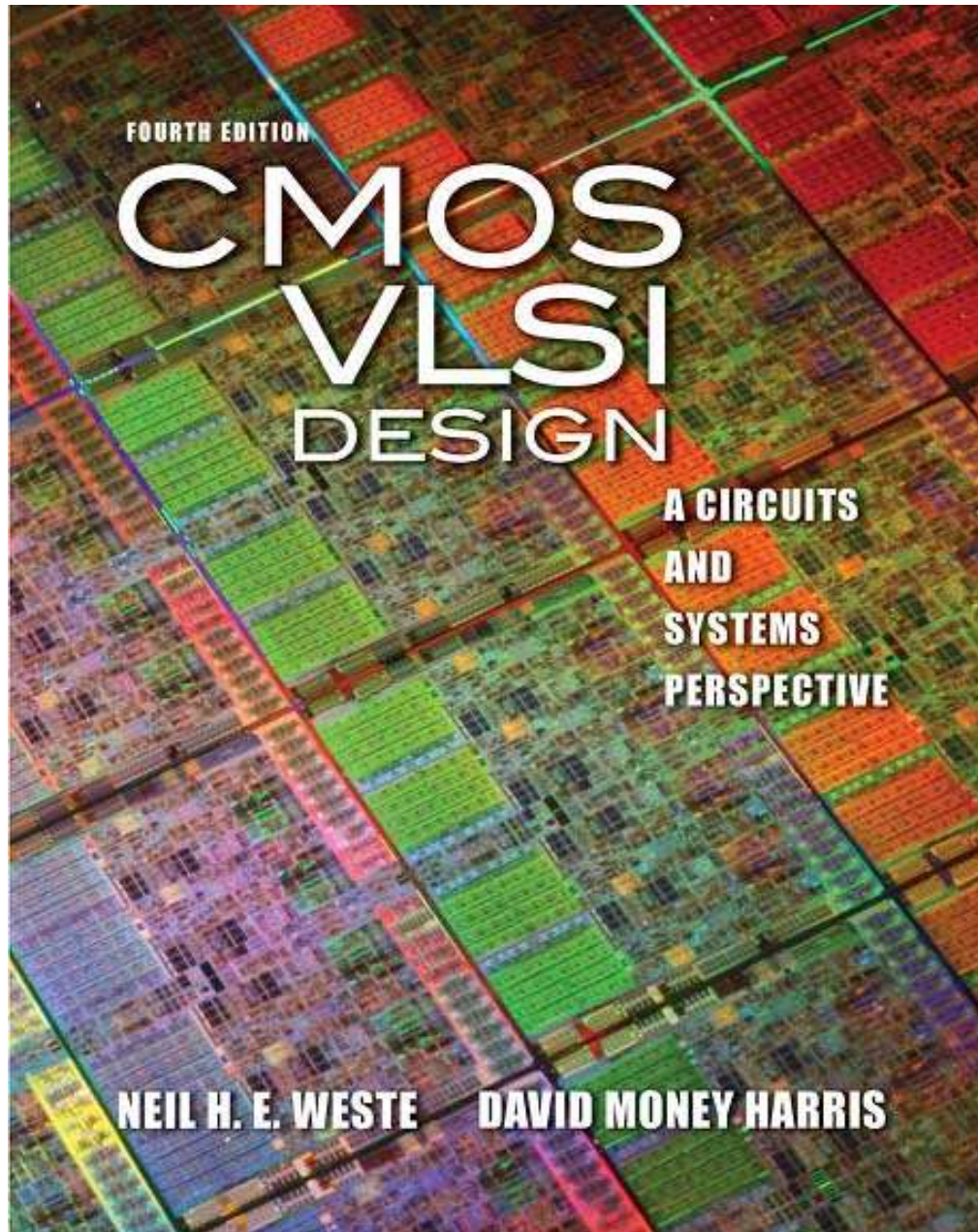




Lecture 6: Circuit design – part 1

- 6.1 Combinational circuit design**
- 6.2 Sequential circuit design
- 6.3 Circuit simulation
- 6.4. Hardware description language

1. Combinational circuit design

- ❑ Circuit designer must learn to think in terms of NAND and NOR to take advantage of static CMOS
- ❑ CMOS stages are inherently inverting, DeMorgan's law helps with this conversion

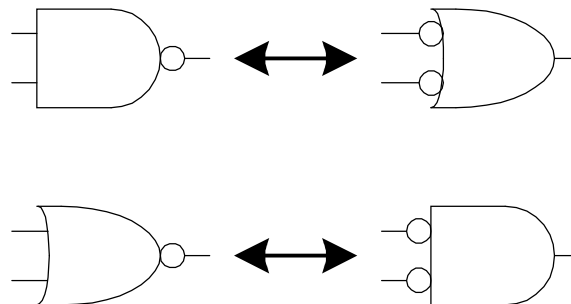
$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$

1. Combinational circuit design

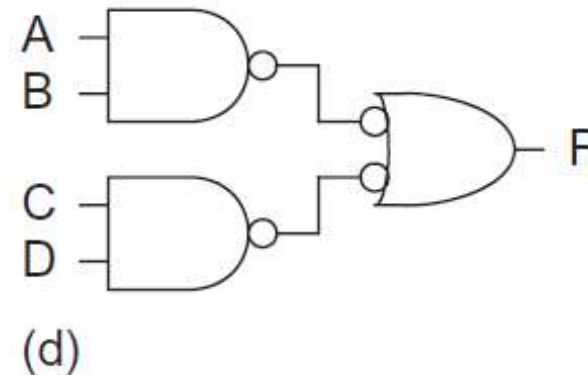
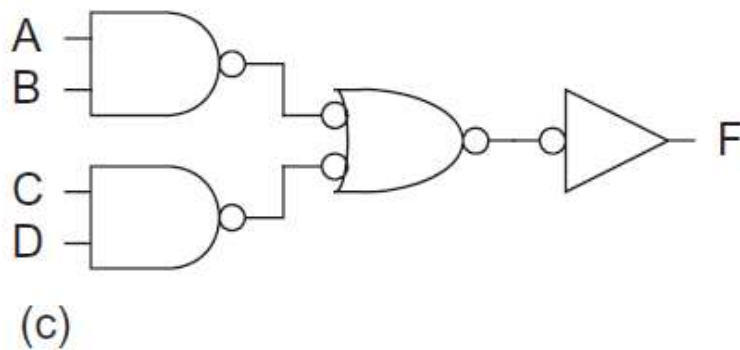
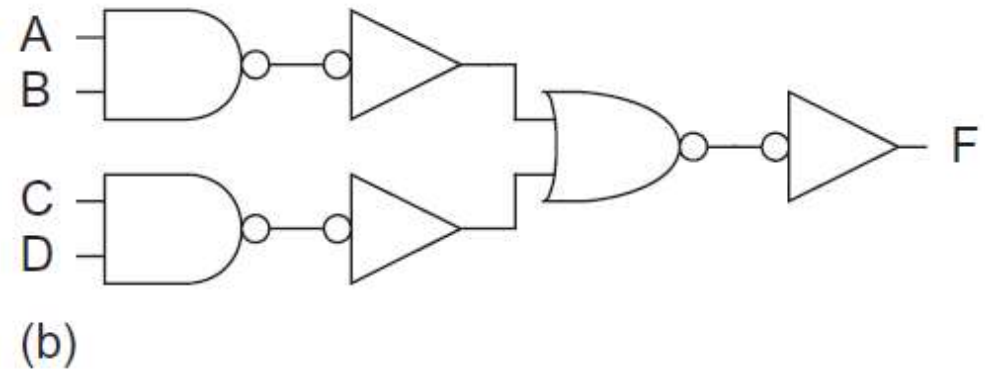
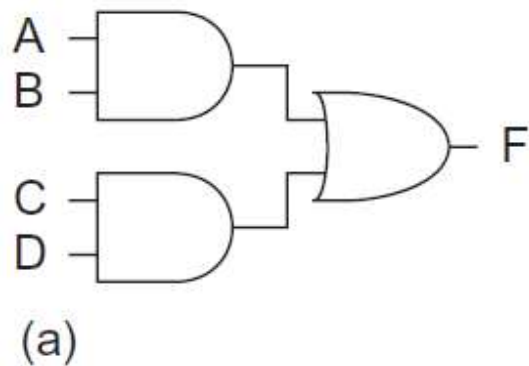
□ Bubble Pushing

- Start with network of AND / OR gates
- Convert to NAND / NOR + inverters
- Push bubbles around to simplify logic
- Remember DeMorgan's Law



Example

- Example: Design a circuit to compute $F = AB + CD$ using NANDs and NORs.

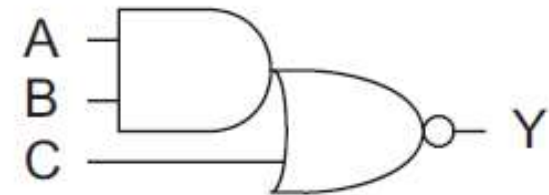


Compound Gates

□ Compound Gates

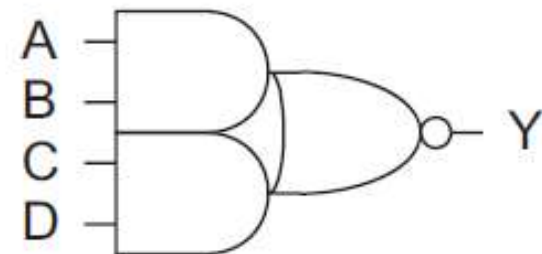
– AOI21: AND-OR-INVERT-21

$$Y = \overline{A \cdot B + C}$$



– AOI22: AND-OR-INVERT-22

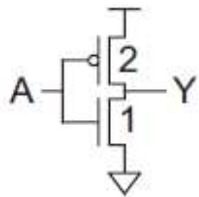
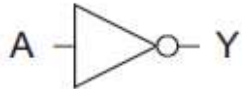
$$Y = \overline{A \cdot B + C \cdot D}$$



Logical Effort of compound gates

Unit Inverter

$$Y = \overline{A}$$

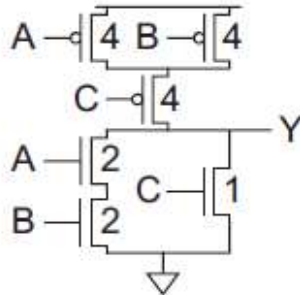
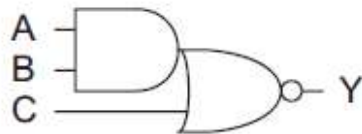


$$g_A = 3/3$$

$$p = 3/3$$

AOI21

$$Y = \overline{A \cdot B + C}$$



$$g_A = 6/3$$

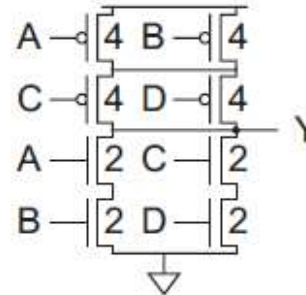
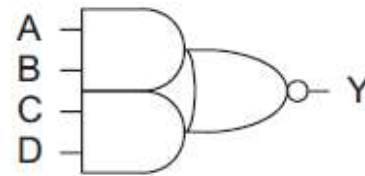
$$g_B = 6/3$$

$$g_C = 5/3$$

$$p = 7/3$$

AOI22

$$Y = \overline{A \cdot B + C \cdot D}$$



$$g_A = 6/3$$

$$g_B = 6/3$$

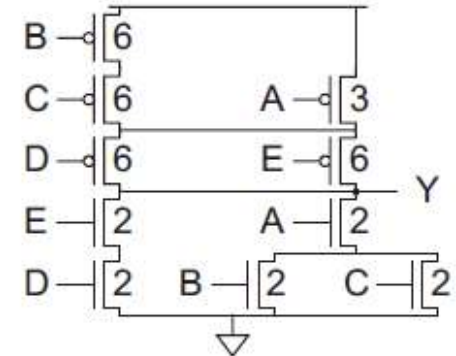
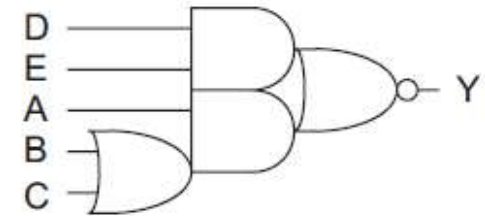
$$g_C = 6/3$$

$$g_D = 6/3$$

$$p = 12/3$$

Complex AOI

$$Y = \overline{A \cdot (B + C) + D \cdot E}$$



$$g_A = 5/3$$

$$g_B = 8/3$$

$$g_C = 8/3$$

$$g_D = 8/3$$

$$g_E = 8/3$$

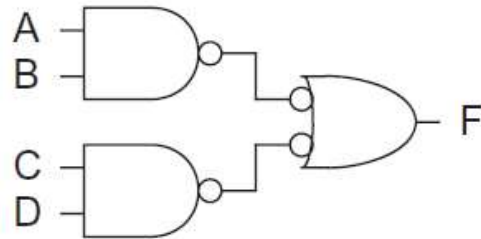
$$p = 16/3$$

Example

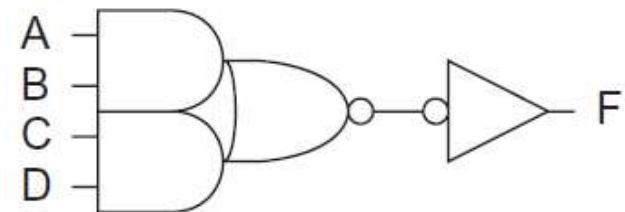
□ Example:

- Calculate the minimum delay, in τ , to compute $F = AB + CD$ using the following circuits:

Using
NAND
gate



Using
compound
gate



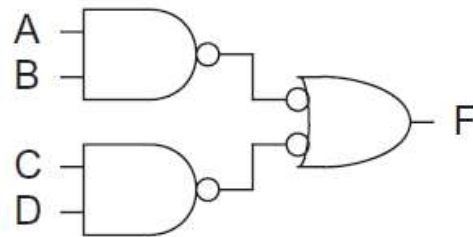
- Each input can present a maximum of 20λ of transistor width. The output must drive a load equivalent to 100λ of transistor width.

Example

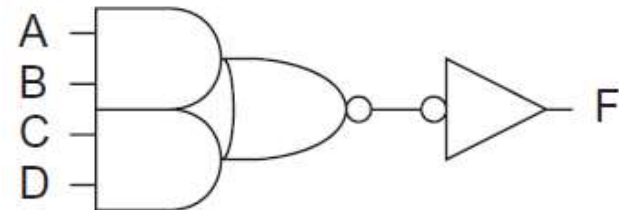
□ Solution:

- The path electrical effort is $H = 100/20 = 5$
- The branching effort is $B = 1$

Using
NAND
gate



Using
compound
gate



No. of stages $N = 2$

Logical effort $G = (4/3) \times (4/3) = 16/9$

Parasitic delay $P = 2 + 2 = 4$

Path efforts $F = GBH = 80/9$

Path delays $D = NF^{1/N} + P = 9.96$

$N = 2$

$G = (6/3) \times 1 = 2$

$P = 12/3 + 1 = 5$

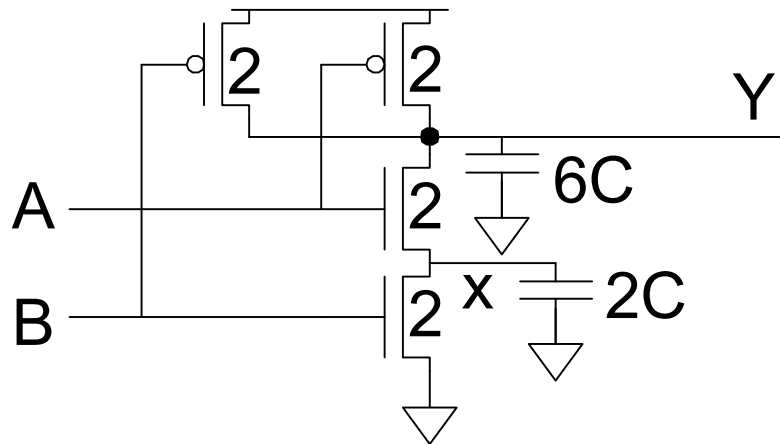
$F = GBH = 2 \times 1 \times 5 = 10$

$D = NF^{1/N} + P = 11.32$

=> Using compound gates does not always result in faster circuits;
simple 2-input NAND gates can be quite fast.

Input Order

- ❑ Our parasitic delay model was too simple
 - Calculate parasitic delay for Y falling
 - If A arrives latest?
 - If B arrives latest?



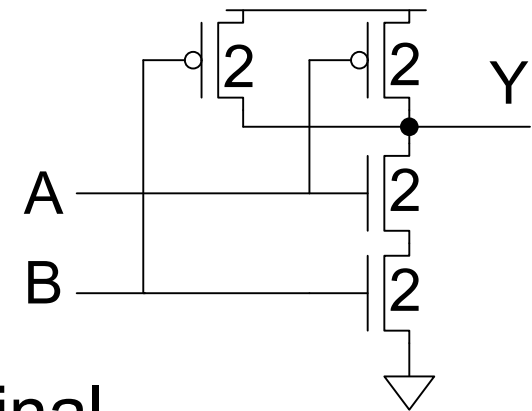
$$D_A = R(6C) = 6RC = 2 \tau$$

$$D_B = (R/2)(2C) + R(6C) = 7RC = 2.33 \tau$$

Inner & Outer Inputs

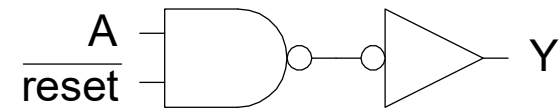
- ❑ *Inner* input is closest to output (A)
- ❑ *Outer* input is closest to rail (B)

- ❑ If input arrival time is known
 - Connect latest input to inner terminal

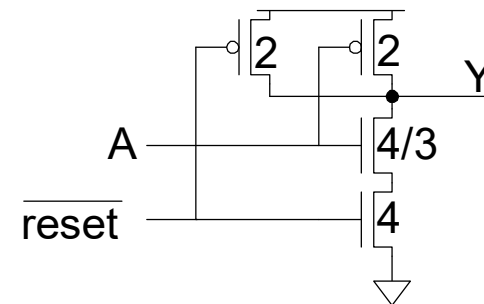


Asymmetric Gates

- ❑ Asymmetric gates favor one input over another
- ❑ Ex: suppose input A of a NAND gate is most critical
 - Use smaller transistor on A (less capacitance)
 - Boost size of noncritical input
 - So total resistance is same

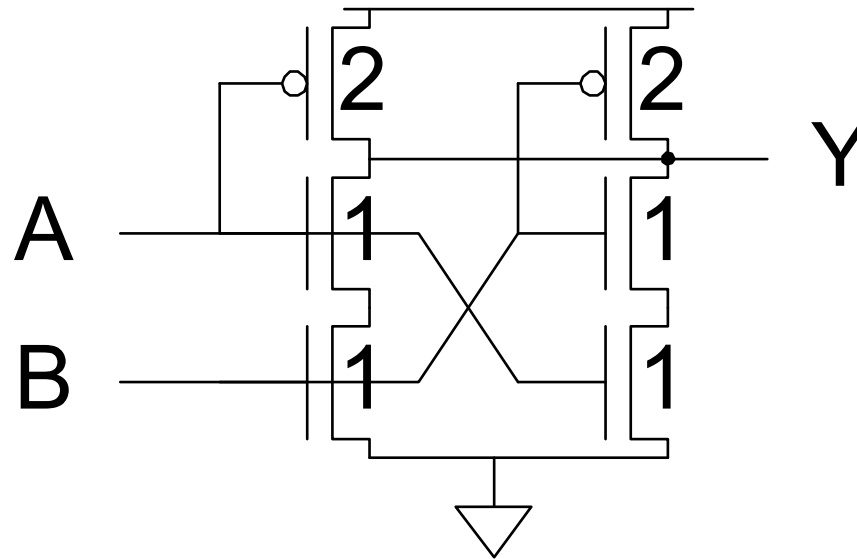


- ❑ $g_A =$
- ❑ $g_B =$
- ❑ $g_{\text{total}} = g_A + g_B =$
- ❑ Asymmetric gate approaches $g = 1$ on critical input
- ❑ But total logical effort goes up



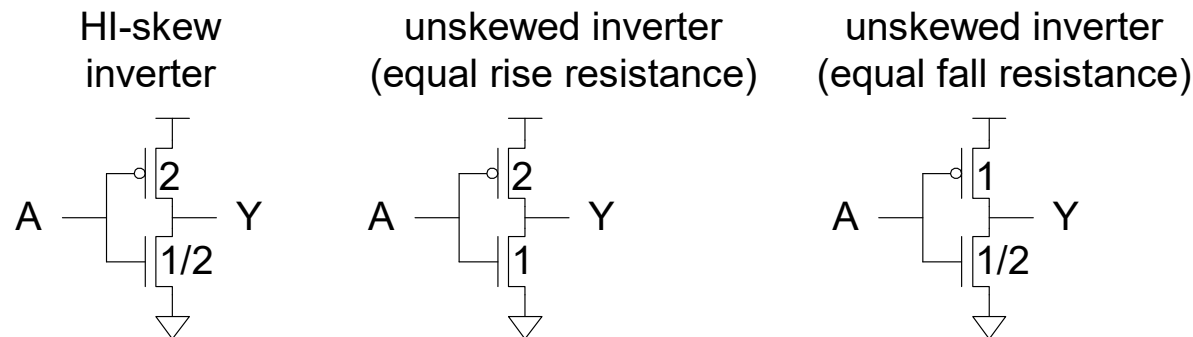
Symmetric Gates

- Inputs can be made perfectly symmetric



Skewed Gates

- ❑ Skewed gates favor one edge over another
- ❑ Ex: suppose rising output of inverter is most critical
 - Downsize noncritical nMOS transistor

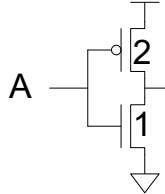
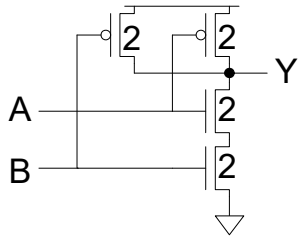
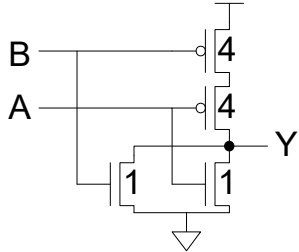
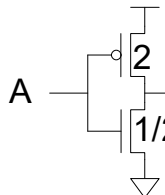
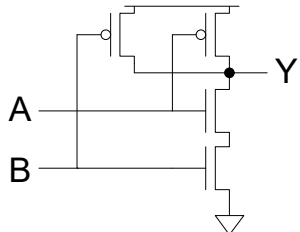
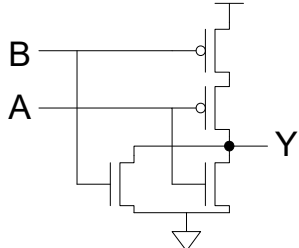
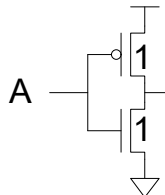
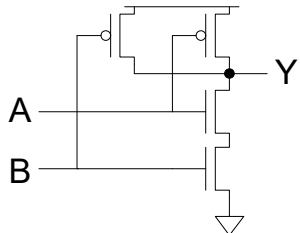
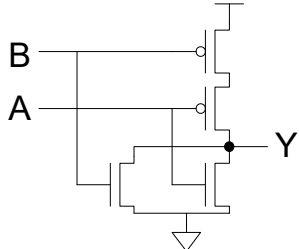


- ❑ Calculate logical effort by comparing to unskewed inverter with same effective resistance on that edge.
 - $g_u =$
 - $g_d =$

HI- and LO-Skew

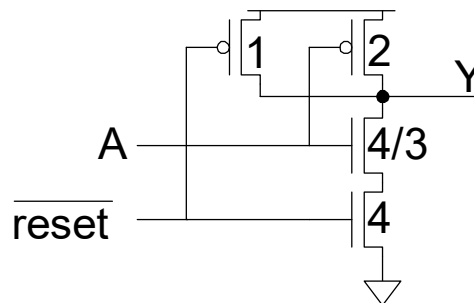
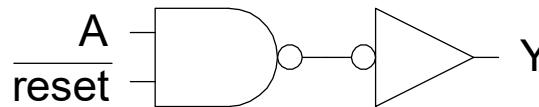
- ❑ Def: Logical effort of a skewed gate for a particular transition is the ratio of the input capacitance of that gate to the input capacitance of an unskewed inverter delivering the same output current for the same transition.
- ❑ Skewed gates reduce size of noncritical transistors
 - HI-skew gates favor rising output (small nMOS)
 - LO-skew gates favor falling output (small pMOS)
- ❑ Logical effort is smaller for favored direction
- ❑ But larger for the other direction

Catalog of Skewed Gates

	Inverter	NAND2	NOR2
unskewed	 $\begin{aligned} g_u &= 1 \\ g_d &= 1 \\ g_{avg} &= 1 \end{aligned}$	 $\begin{aligned} g_u &= 4/3 \\ g_d &= 4/3 \\ g_{avg} &= 4/3 \end{aligned}$	 $\begin{aligned} g_u &= 5/3 \\ g_d &= 5/3 \\ g_{avg} &= 5/3 \end{aligned}$
HI-skew	 $\begin{aligned} g_u &= 5/6 \\ g_d &= 5/3 \\ g_{avg} &= 5/4 \end{aligned}$	 $\begin{aligned} g_u &= \\ g_d &= \\ g_{avg} &= \end{aligned}$	 $\begin{aligned} g_u &= \\ g_d &= \\ g_{avg} &= \end{aligned}$
LO-skew	 $\begin{aligned} g_u &= 4/3 \\ g_d &= 2/3 \\ g_{avg} &= 1 \end{aligned}$	 $\begin{aligned} g_u &= \\ g_d &= \\ g_{avg} &= \end{aligned}$	 $\begin{aligned} g_u &= \\ g_d &= \\ g_{avg} &= \end{aligned}$

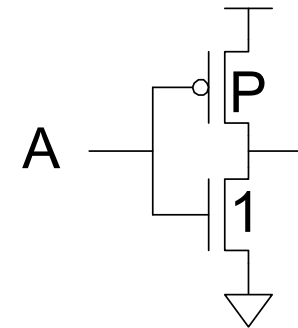
Asymmetric Skew

- ❑ Combine asymmetric and skewed gates
 - Downsize noncritical transistor on unimportant input
 - Reduces parasitic delay for critical input



Best P/N Ratio

- ❑ We have selected P/N ratio for unit rise and fall resistance ($\mu = 2-3$ for an inverter).
- ❑ Alternative: choose ratio for least average delay
- ❑ Ex: inverter
 - Delay driving identical inverter
 - $t_{pdf} =$
 - $t_{pdr} =$
 - $t_{pd} =$
 - $dt_{pd} / dP =$
 - Least delay for $P =$

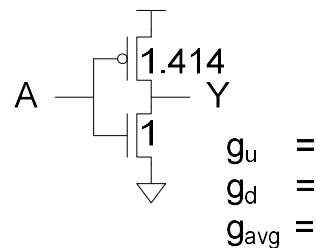


P/N Ratios

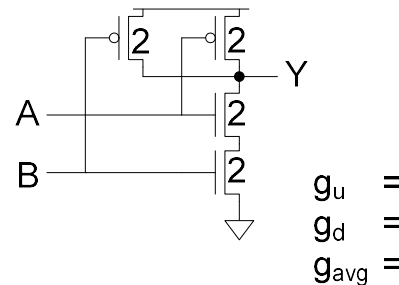
- ❑ In general, best P/N ratio is sqrt of equal delay ratio.
 - Only improves average delay slightly for inverters
 - But significantly decreases area and power

fastest
P/N ratio

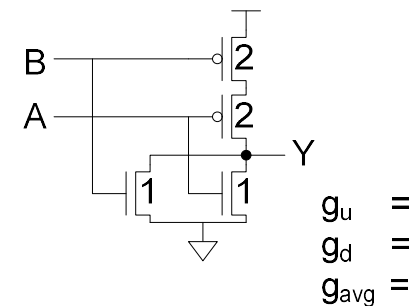
Inverter



NAND2



NOR2



Observations

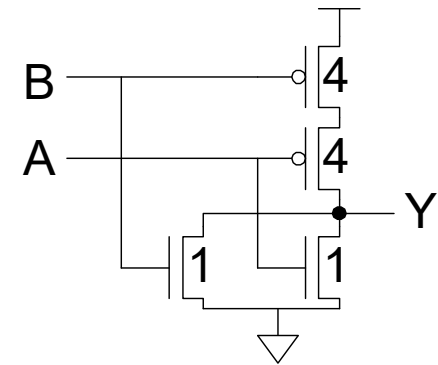
- ❑ For speed:
 - NAND vs. NOR
 - Many simple stages vs. fewer high fan-in stages
 - Latest-arriving input
- ❑ For area and power:
 - Many simple stages vs. fewer high fan-in stages

Review

1. What are compound gates?
2. What are *inner* input and *outer* input?
3. When do we need asymmetric gates?
4. What are HI-skew and LOW-skew?
5. What is the best P/N ratio?
6. Calculate the logical effort and parasitic delays of the following AOI gates:
 1. $Y = \overline{ABC + DE}$
 2. $Y = \overline{A + BC + DE}$

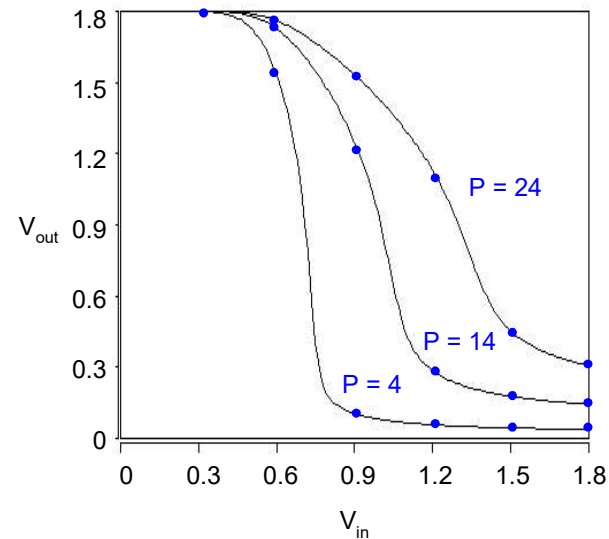
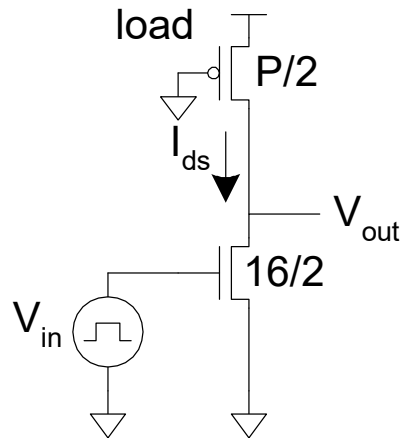
Pseudo-nMOS

- ❑ What makes a circuit fast?
 - $I = C \, dV/dt \rightarrow t_{pd} \propto (C/I) \Delta V$
 - low capacitance
 - high current
 - small swing
- ❑ Logical effort is proportional to C/I
- ❑ pMOS are the enemy!
 - High capacitance for a given current
- ❑ Can we take the pMOS capacitance off the input?
- ❑ Various circuit families try to do this...



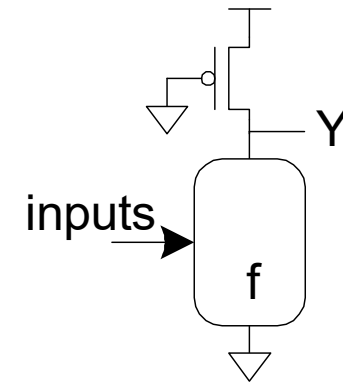
Pseudo-nMOS

- ❑ In the old days, nMOS processes had no pMOS
 - Instead, use pull-up transistor that is always ON
- ❑ In CMOS, use a pMOS that is always ON
 - *Ratio* issue
 - Make pMOS about $\frac{1}{4}$ effective strength of pulldown network

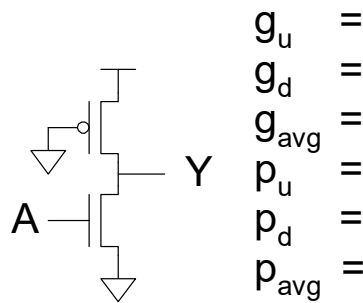


Pseudo-nMOS Gates

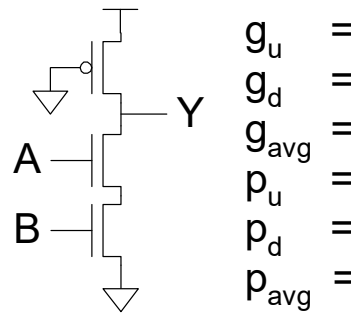
- ❑ Design for unit current on output to compare with unit inverter.
- ❑ pMOS fights nMOS



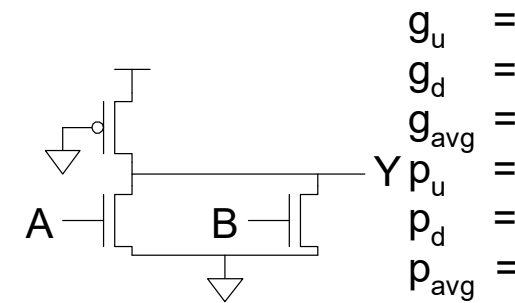
Inverter



NAND2

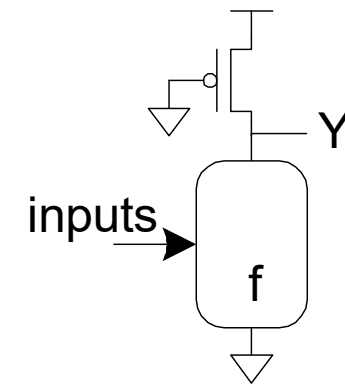


NOR2

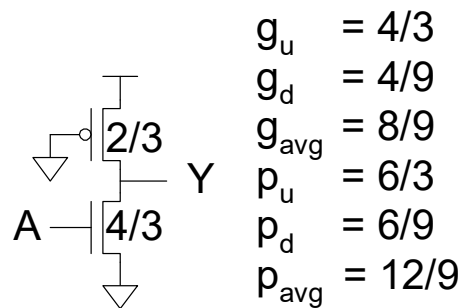


Pseudo-nMOS Gates

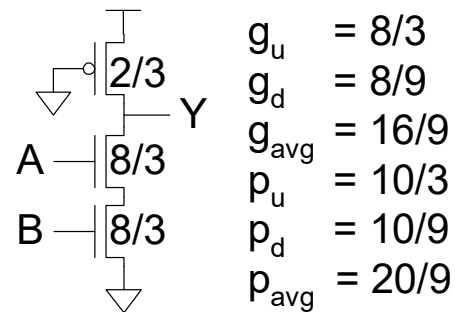
- ❑ Design for unit current on output to compare with unit inverter.
- ❑ pMOS fights nMOS



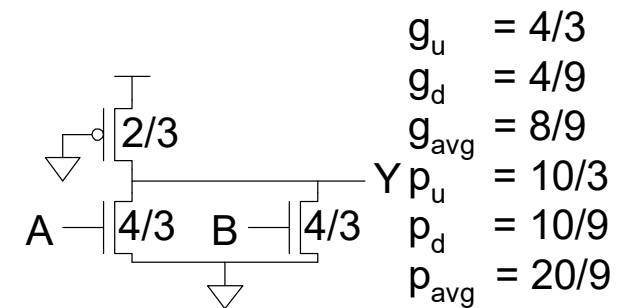
Inverter



NAND2



NOR2



Pseudo-nMOS Design

- Ex: Design a k-input AND gate using pseudo-nMOS. Estimate the delay driving a fanout of H

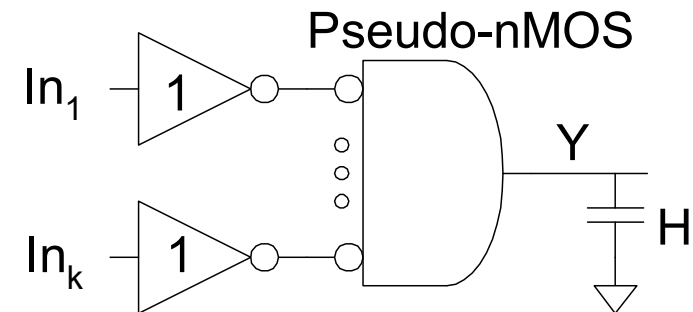
- $G =$

- $F =$

- $P =$

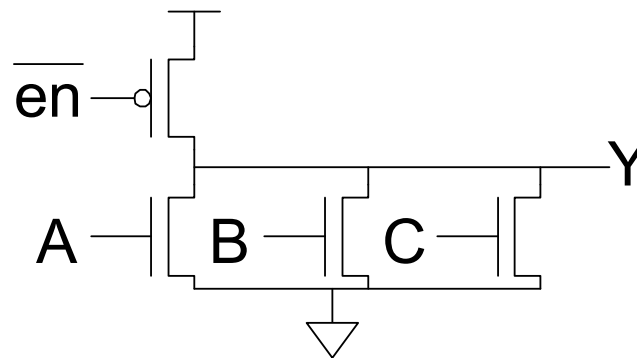
- $N =$

- $D =$



Pseudo-nMOS Power

- ❑ Pseudo-nMOS draws power whenever $Y = 0$
 - Called static power $P = I_{DD} V_{DD}$
 - A few mA / gate * 1M gates would be a problem
 - Explains why nMOS went extinct
- ❑ Use pseudo-nMOS sparingly for wide NORs
- ❑ Turn off pMOS when not in use



Ratio Example

- ❑ The chip contains a 32 word x 48 bit ROM
 - Uses pseudo-nMOS decoder and bitline pullups
 - On average, one wordline and 24 bitlines are high
- ❑ Find static power drawn by the ROM
 - $I_{\text{on-p}} = 36 \mu\text{A}$, $V_{\text{DD}} = 1.0 \text{ V}$

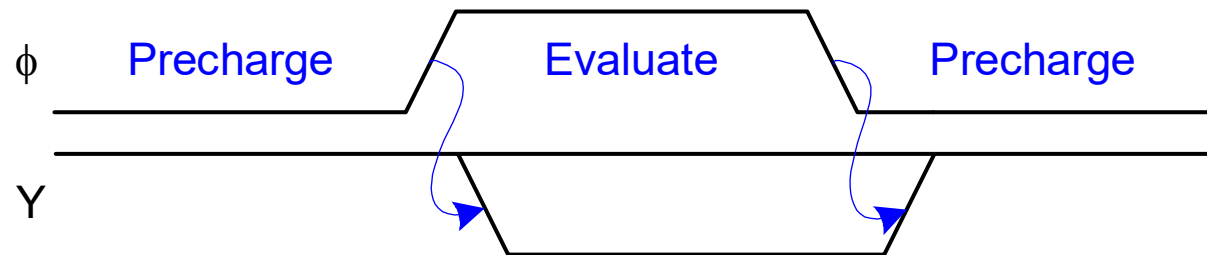
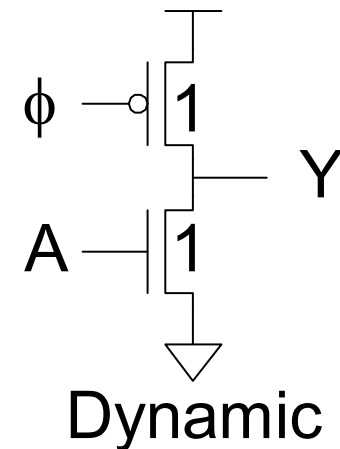
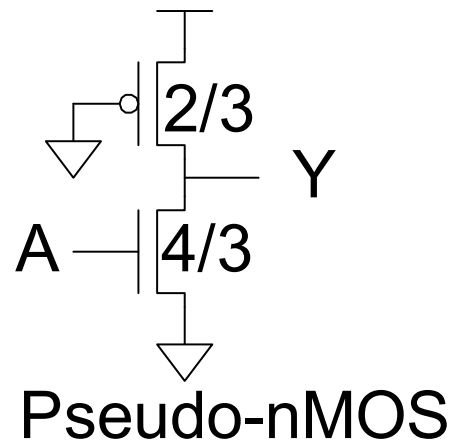
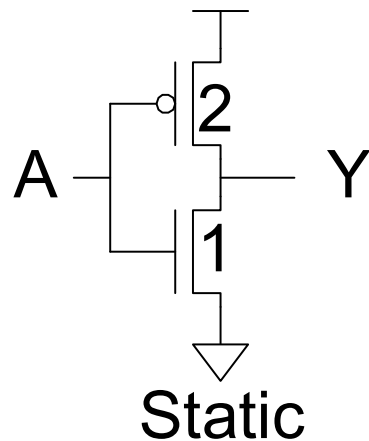
❑ Solution:

$$P_{\text{pull-up}} =$$

$$P_{\text{static}} =$$

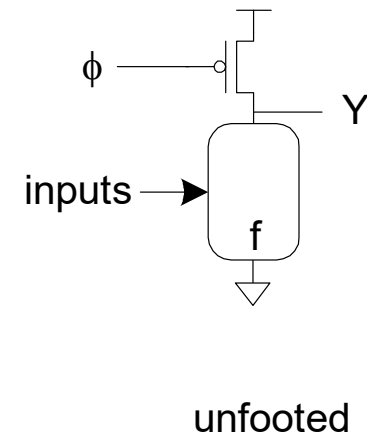
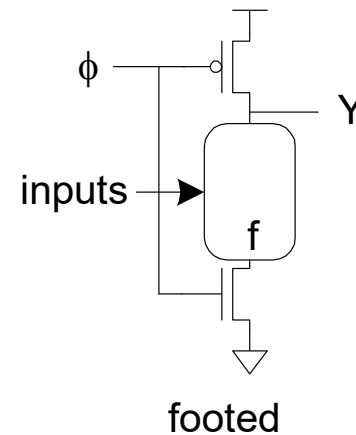
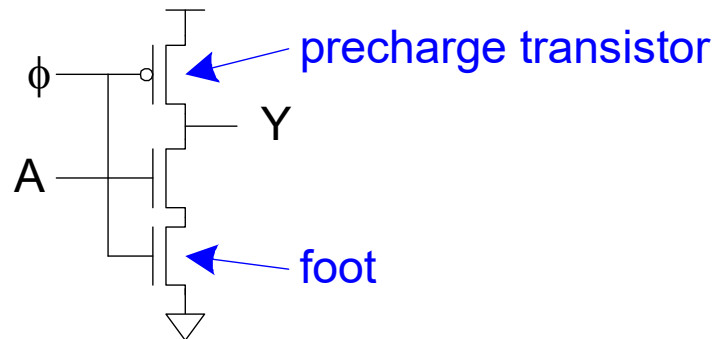
Dynamic Logic

- ❑ *Dynamic* gates uses a clocked pMOS pullup
- ❑ Two modes: *precharge* and *evaluate*



The Foot

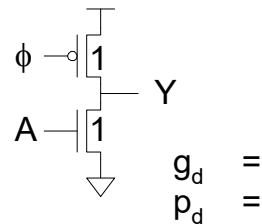
- ❑ What if pulldown network is ON during precharge?
- ❑ Use series evaluation transistor to prevent fight.



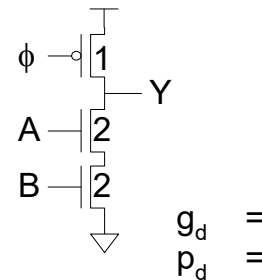
Logical Effort

unfooted

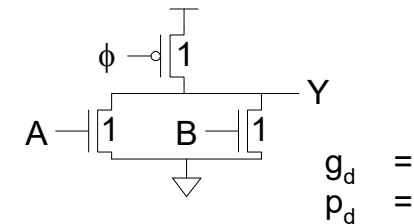
Inverter



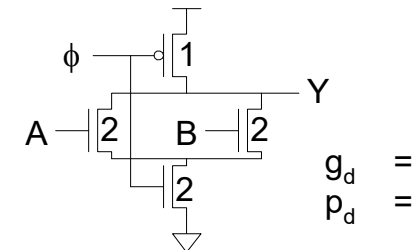
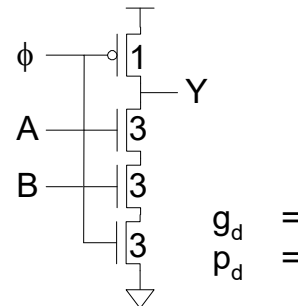
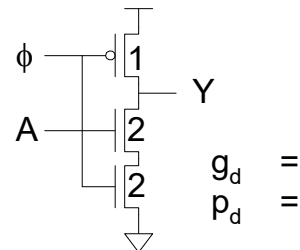
NAND2



NOR2



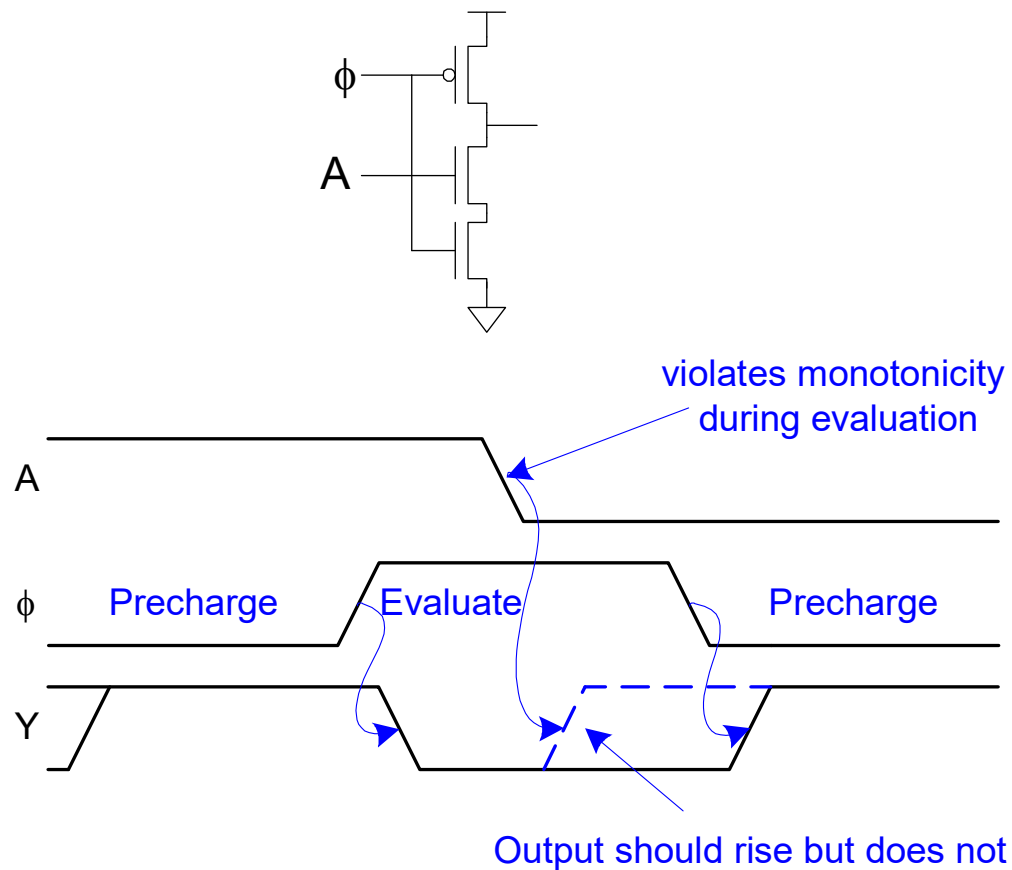
footed



Monotonicity

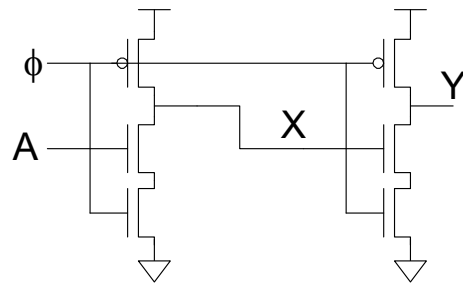
❑ Dynamic gates require *monotonically rising* inputs during evaluation

- 0 $\rightarrow$ 0
- 0 $\rightarrow$ 1
- 1 $\rightarrow$ 1
- But not 1 $\rightarrow$ 0

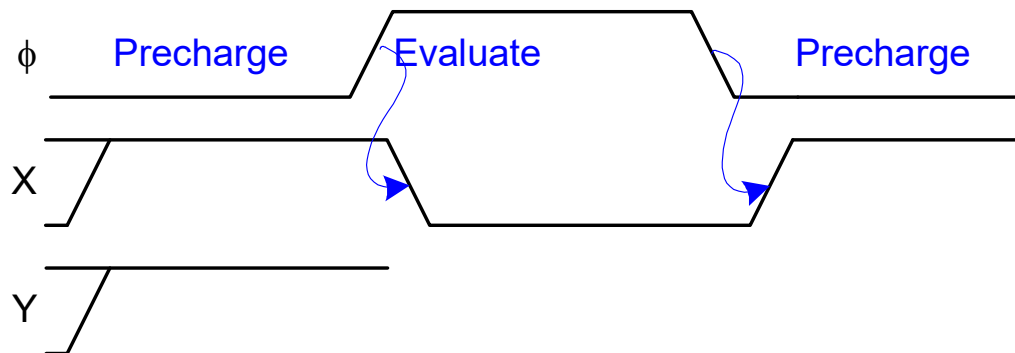


Monotonicity Woes

- ❑ But dynamic gates produce monotonically falling outputs during evaluation
- ❑ Illegal for one dynamic gate to drive another!

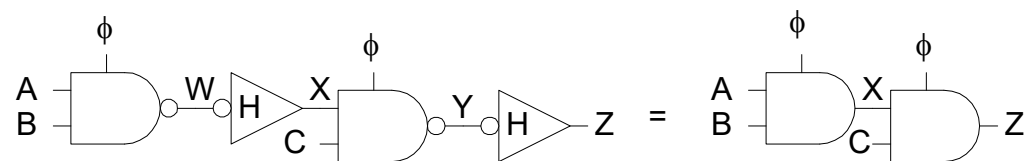
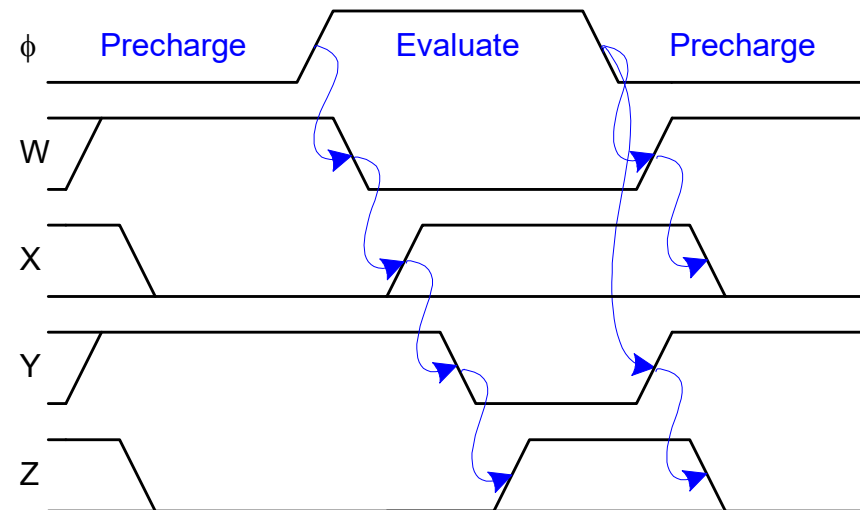
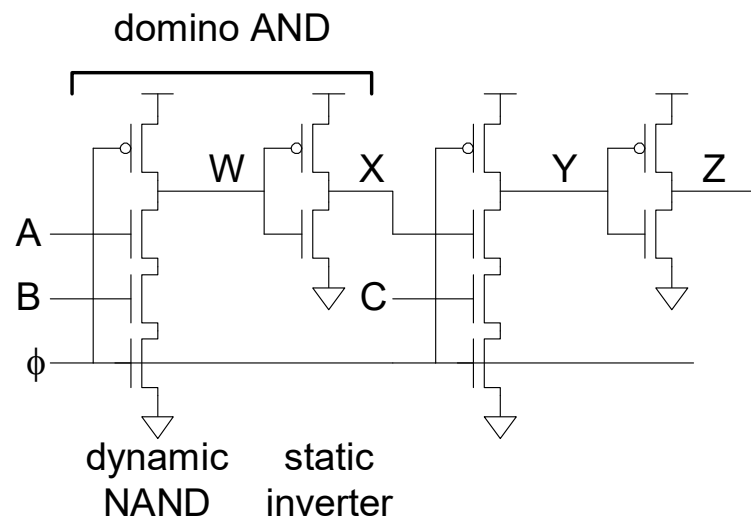


A = 1



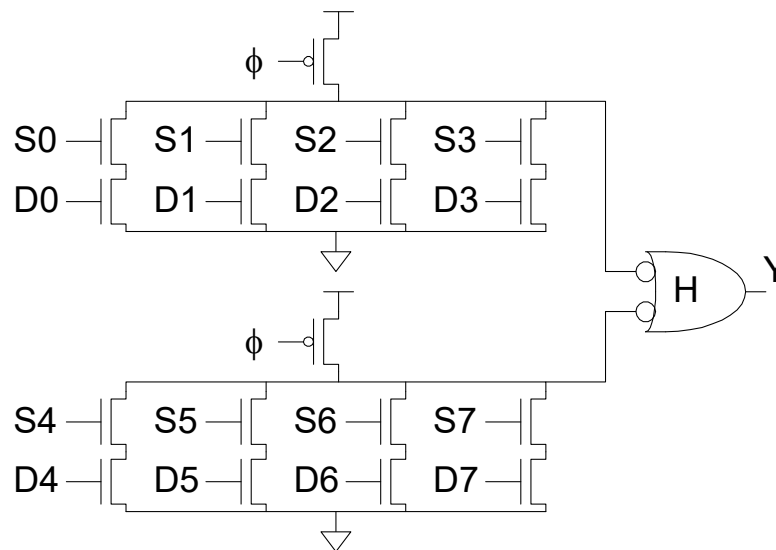
Domino Gates

- ❑ Follow dynamic stage with inverting static gate
 - Dynamic / static pair is called domino gate
 - Produces monotonic outputs



Domino Optimizations

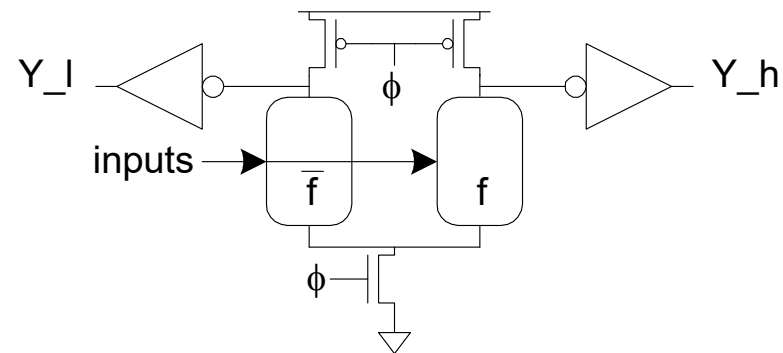
- ❑ Each domino gate triggers next one, like a string of dominos toppling over
- ❑ Gates evaluate sequentially but precharge in parallel
- ❑ Thus evaluation is more critical than precharge
- ❑ HI-skewed static stages can perform logic



Dual-Rail Domino

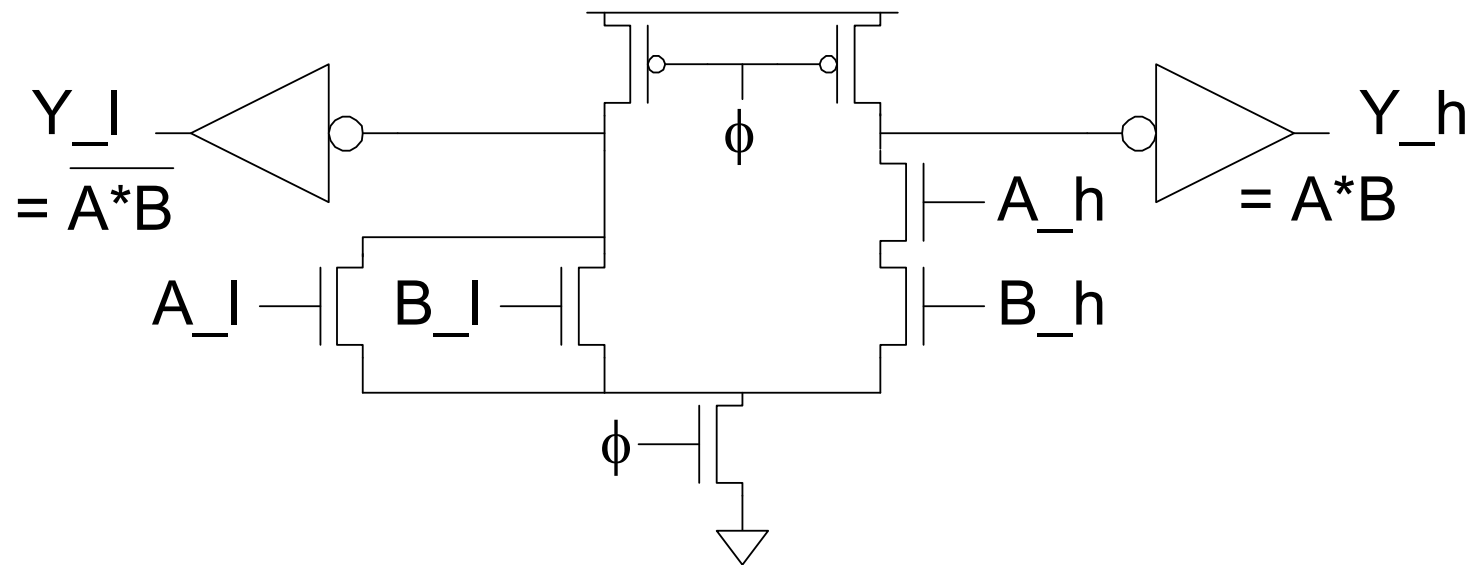
- ❑ Domino only performs noninverting functions:
 - AND, OR but not NAND, NOR, or XOR
- ❑ Dual-rail domino solves this problem
 - Takes true and complementary inputs
 - Produces true and complementary outputs

sig_h	sig_l	Meaning
0	0	Precharged
0	1	'0'
1	0	'1'
1	1	invalid



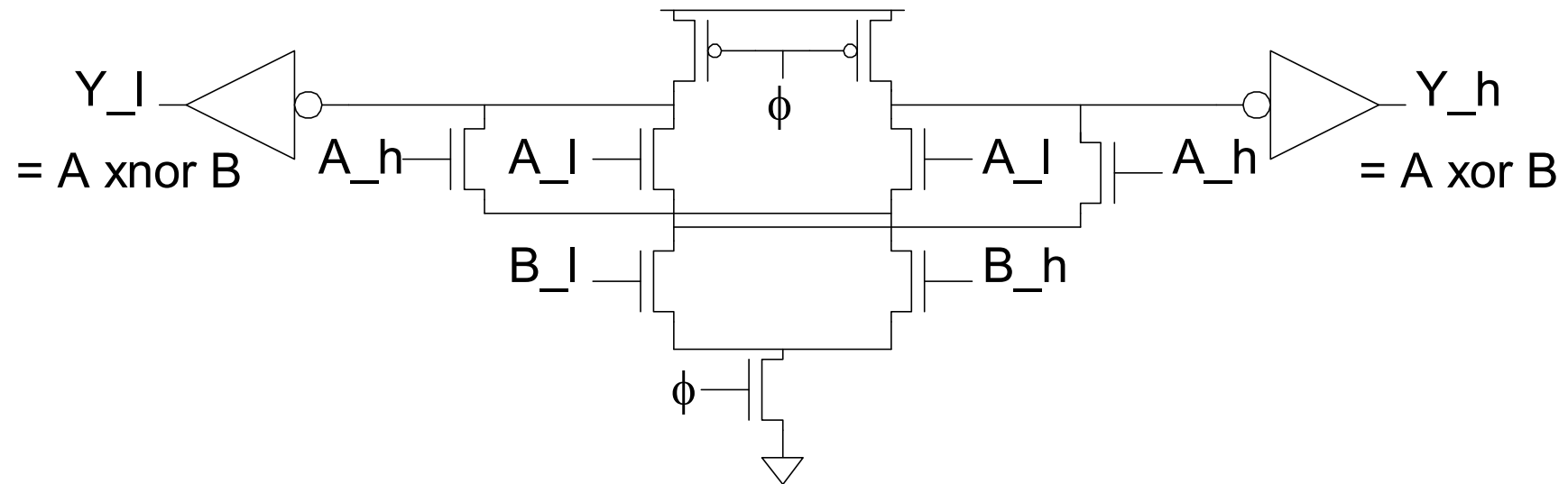
Example: AND/NAND

- ❑ Given A_h, A_l, B_h, B_l
- ❑ Compute $Y_h = AB, Y_l = \overline{AB}$
- ❑ Pulldown networks are conduction complements



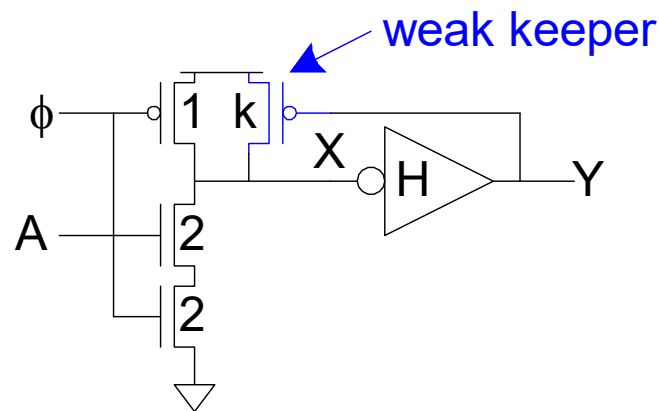
Example: XOR/XNOR

- Sometimes possible to share transistors



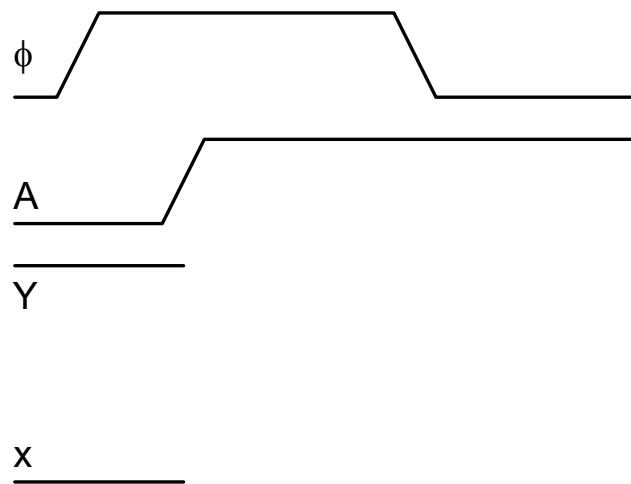
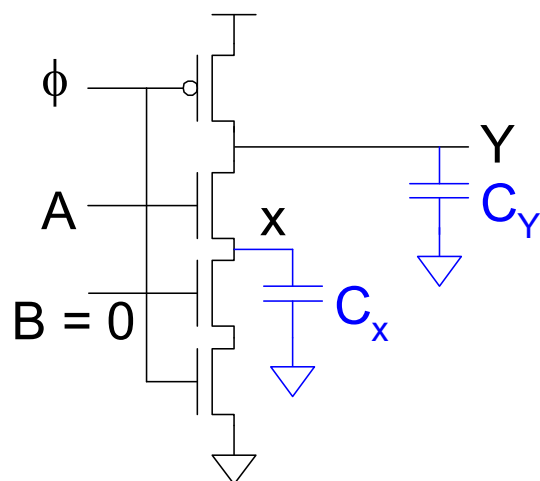
Leakage

- ❑ Dynamic node floats high during evaluation
 - Transistors are leaky ($I_{OFF} \neq 0$)
 - Dynamic value will leak away over time
 - Formerly milliseconds, now nanoseconds
- ❑ Use keeper to hold dynamic node
 - Must be weak enough not to fight evaluation



Charge Sharing

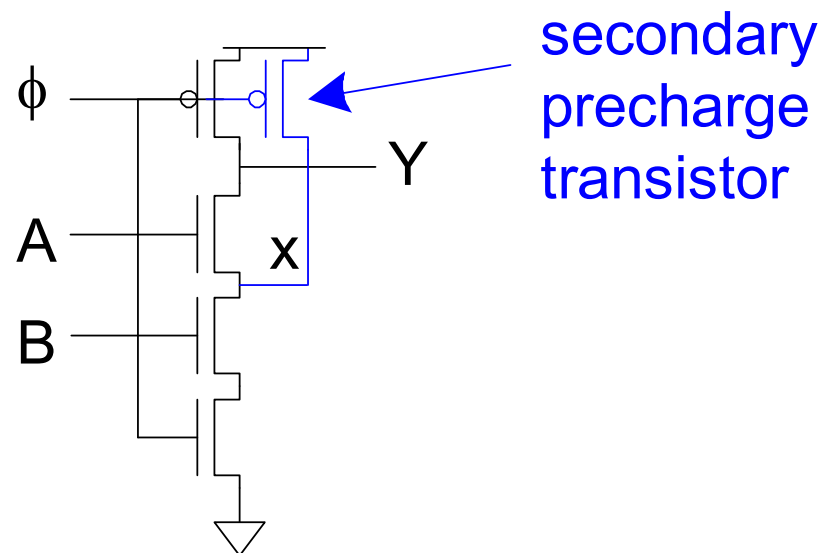
- ❑ Dynamic gates suffer from charge sharing



$$V_x = V_Y =$$

Secondary Precharge

- ❑ Solution: add secondary precharge transistors
 - Typically need to precharge every other node
- ❑ Big load capacitance C_Y helps as well



Noise Sensitivity

- ❑ Dynamic gates are very sensitive to noise
 - Inputs: $V_{IH} \approx V_{tn}$
 - Outputs: floating output susceptible noise
- ❑ Noise sources
 - Capacitive crosstalk
 - Charge sharing
 - Power supply noise
 - Feedthrough noise
 - And more!

Power

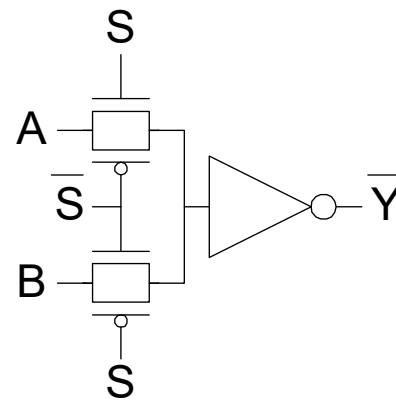
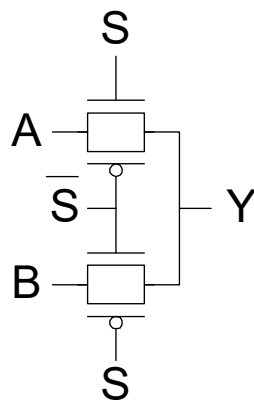
- ❑ Domino gates have high activity factors
 - Output evaluates and precharges
 - If output probability = 0.5, $\alpha = 0.5$
 - Output rises and falls on half the cycles
 - Clocked transistors have $\alpha = 1$
- ❑ Leads to very high power consumption

Domino Summary

- ❑ Domino logic is attractive for high-speed circuits
 - 1.3 – 2x faster than static CMOS
 - But many challenges:
 - Monotonicity, leakage, charge sharing, noise
- ❑ Widely used in high-performance microprocessors in 1990s when speed was king
- ❑ Largely displaced by static CMOS now that power is the limiter
- ❑ Still used in memories for area efficiency

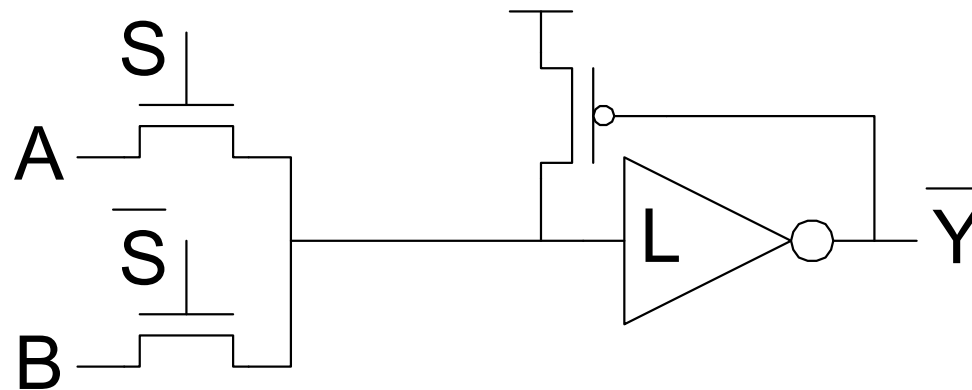
Pass Transistor Circuits

- ❑ Use pass transistors like switches to do logic
- ❑ Inputs drive diffusion terminals as well as gates
- ❑ CMOS + Transmission Gates:
 - 2-input multiplexer
 - Gates should be restoring



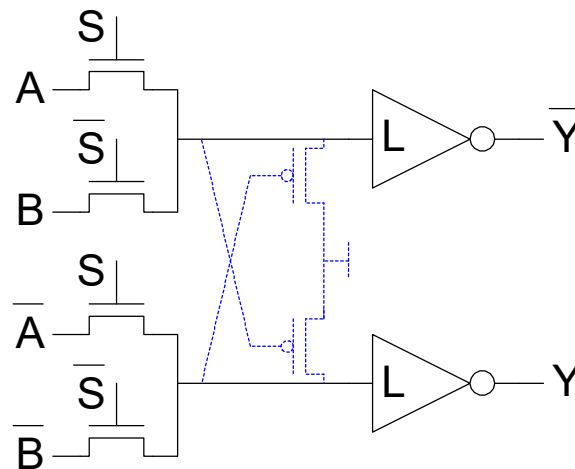
LEAP

- ❑ **LEA**n integration with **P**ass transistors
- ❑ Get rid of pMOS transistors
 - Use weak pMOS feedback to pull fully high
 - Ratio constraint



CPL

- ❑ Complementary Pass-transistor Logic
 - Dual-rail form of pass transistor logic
 - Avoids need for ratioed feedback
 - Optional cross-coupling for rail-to-rail swing



Pass Transistor Summary

- ❑ Researchers investigated pass transistor logic for general purpose applications in the 1990's
 - Benefits over static CMOS were small or negative
 - No longer generally used
- ❑ However, pass transistors still have a niche in special circuits such as memories where they offer small size and the threshold drops can be managed

Review

1. What are pseudo-nMOS gates?
2. What are drawbacks of pseudo-nMOS gates?
3. What are dynamic gates?
4. What is the foot of dynamic gates?
5. Why do we need secondary precharge for dynamic gates?
6. Why are dynamic gates very sensitive to noise?
7. What are domino gates?
8. What is CPL?

Review

1. Compute logical effort for rising and falling g_u and g_d of the following gates

